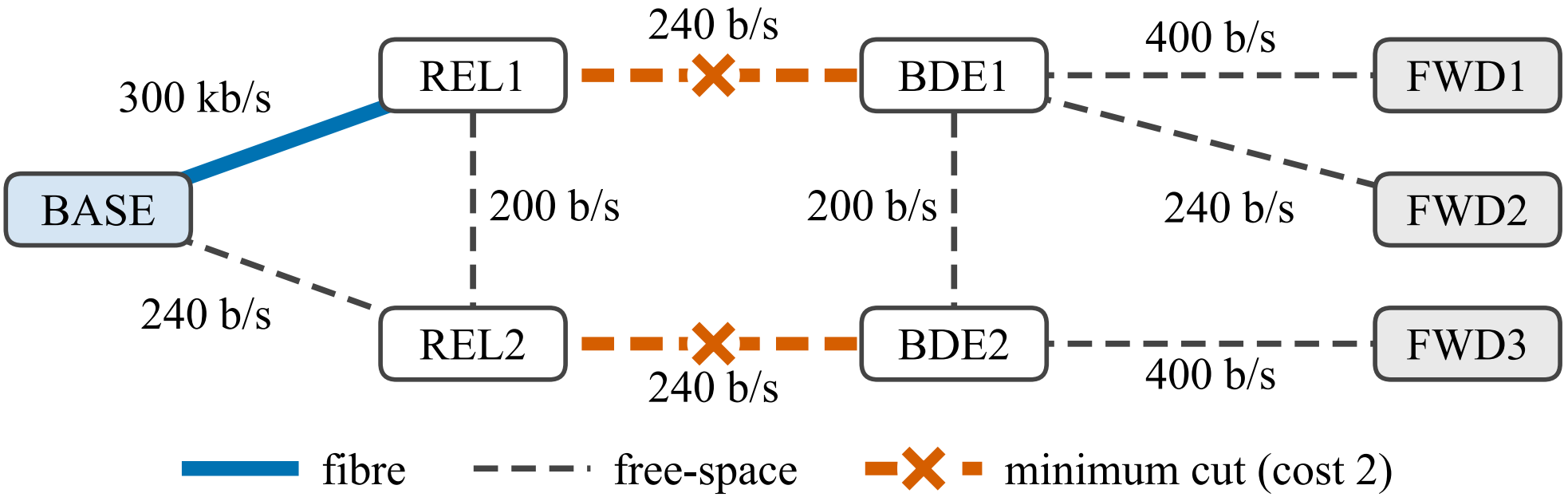
(a) Formation and its two-edge cut



(b) Three failure surfaces, modelled separately

Failure surface	✗ *Stops*	✓ *Still works*
Quantum channel QBER $e_0+\Delta e \geq e_{\text{abort}}$	Fresh key generation **ABORT**	Stored key above reserve; classical relay and fallback
Classical transport denial	Relay and fallback **OUTAGE**	Stored key retained for later use
Online key-establishment denial	KEM fallback setup (G3/G4)	QKD-keyed G1/G2; pre-placed symmetric keys